

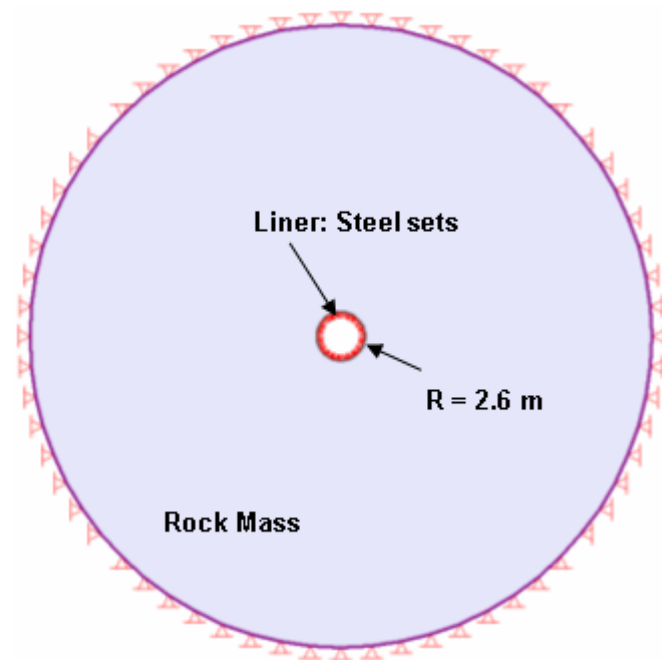
Liner with Sliding Gap

This tutorial describes the support of a tunnel using a circular steel-set liner which includes sliding gaps. The steel-sets are constructed with sliding gaps that enable a squeezing tunnel to contract with little resistance until the gap is closed, at which time the liner will resist further tunnel squeezing. This technique enables the installation of the liner near the tunnel face before large deformations have occurred, but prevents very large forces from building up in the liner since the liner is able to deform a significant amount before it takes axial load.

The analysis will be conducted in two parts. The first part shows the response of the tunnel with a regular liner (no sliding gap). The second part shows the effect of using a liner with sliding gaps.

Topics Covered

- Liners with Sliding Gap
- Simulated three-dimensional tunnel excavation
- Staged tractions
- Liner axial forces
- Graph Single Point vs Stage



Background

Steel sets are sometimes used to support tunnels subjected to high squeezing stresses (see Figure 1). If these supports are installed before much deformation has occurred, the steel sets will be subjected to very large stresses as the tunnel deforms and the supports may fail (see Figure 2). However, it is often necessary to install support as close as possible to the tunnel face to ensure the safety of the workers. For this reason, sliding joints (gaps) may be added to the steel sets (see Figure 3). These gaps allow the liner to easily deform in the axial direction, until a predetermined amount of deformation has occurred, at which time the support will “lock” and will begin resisting axial stresses.

This system allows the installation of the steel sets close to the tunnel face when little tunnel deformation has occurred. The sliding joints will allow further tunnel deformation before the steel sets pick up axial load. This system will prevent extreme deformations in the tunnel but will also prevent failure of the supports by ensuring that they are not subjected to very high stresses.



Figure 1. Steel sets for supporting a 5.2 m diameter tunnel.

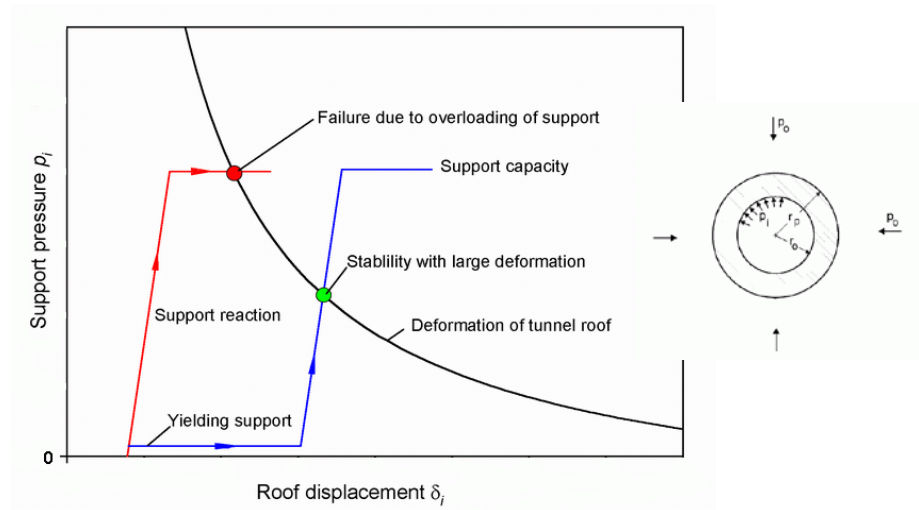


Figure 2. Plot of support pressure versus roof displacement for a squeezing tunnel. The red line shows the pressure on a liner with no sliding joints installed after a small amount of roof displacement. The blue line shows a liner *with* sliding joints installed at the same time.



Figure 3. A sliding joint.

TIP: plots such as Figure 2 can be obtained from the program *RocSupport* available from Rocscience. For more information about *RocSupport* see the Rocscience website.

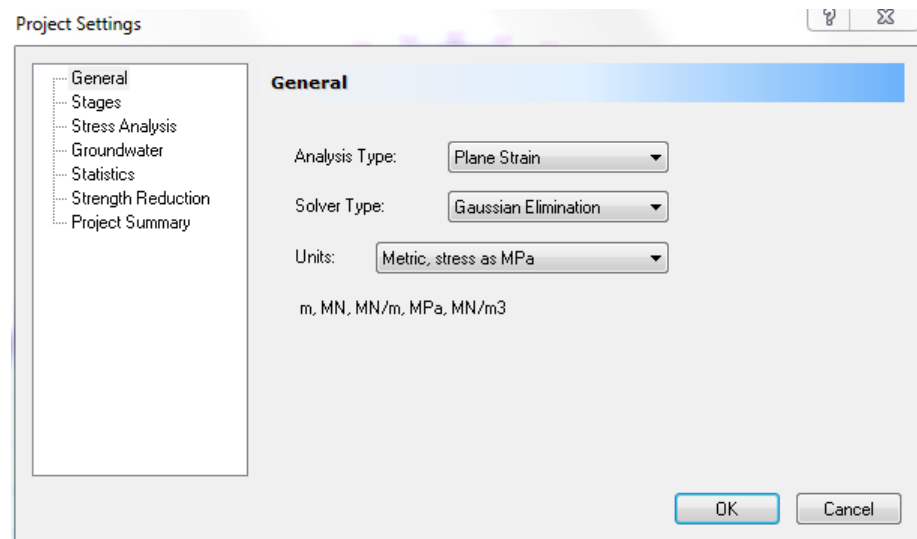
Model

Start the *Phase2* Model program.

Project Settings



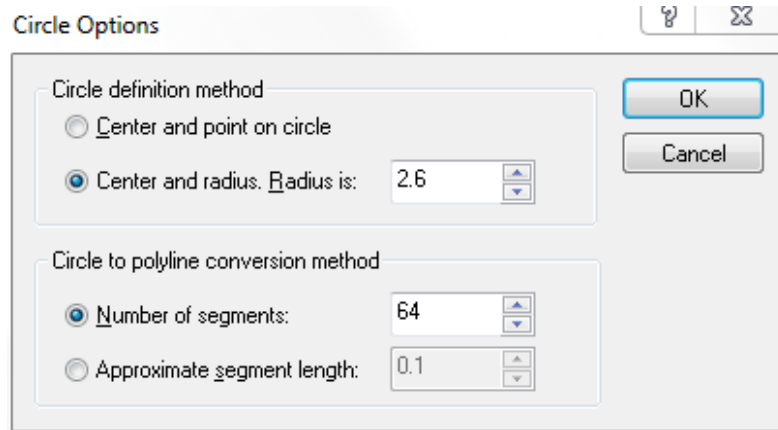
Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Set the Number of Stages to 20. In this analysis, the three-dimensional effects of tunnel excavation will be simulated by gradually decreasing tractions on the surface of the tunnel, therefore many stages are required. Define the units as being “Metric, stress as MPa”. Click OK to close the Project Settings dialog. If you see a warning about the unit system then hit OK.



Boundaries and Mesh



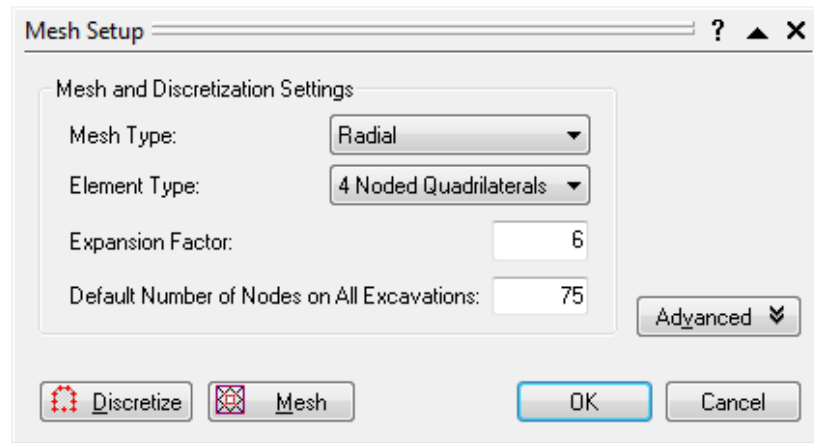
First we will define the circular tunnel. Select **Add Excavation** from the **Boundaries** menu. Type the letter **i** to indicate you wish to draw a circle (or right click and choose Circle) and hit Enter. You will now see the dialog for entering a circle. Select the **Centre and radius** option and set the radius to 2.6. Set the Number of segments to 64 as shown.



Click OK. You will now see a circle that you can drag around with the mouse. Enter 0,0 for the centre coordinates and hit Enter. The excavation geometry is now defined.



The finite element mesh and external boundary can be constructed at the same time by generating a Radial mesh. Select **Mesh Setup** from the **Mesh** menu. For Mesh Type choose Radial, for Element Type choose 4 Noded Quadrilaterals and for Expansion Factor enter 6. Click the Discretize button and then the Mesh button to generate the mesh and the external boundary.



Click OK to close the dialog. Your model should appear as shown below. Note the default boundary conditions for the external boundary are fixed (zero displacement), which is what we want.

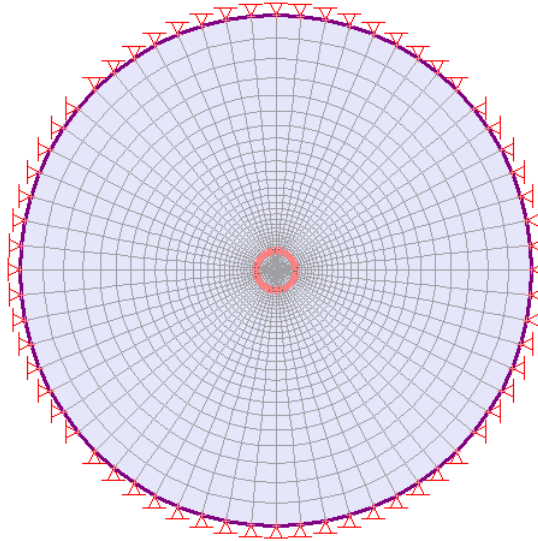


Figure 4. Circular tunnel with radial mesh.

Field Stress



Select **Field Stress** from the **Loading** menu. For this model we will assume a constant hydrostatic stress. Enter a value of 30 MPa for σ_1 , σ_3 and σ_z as shown.

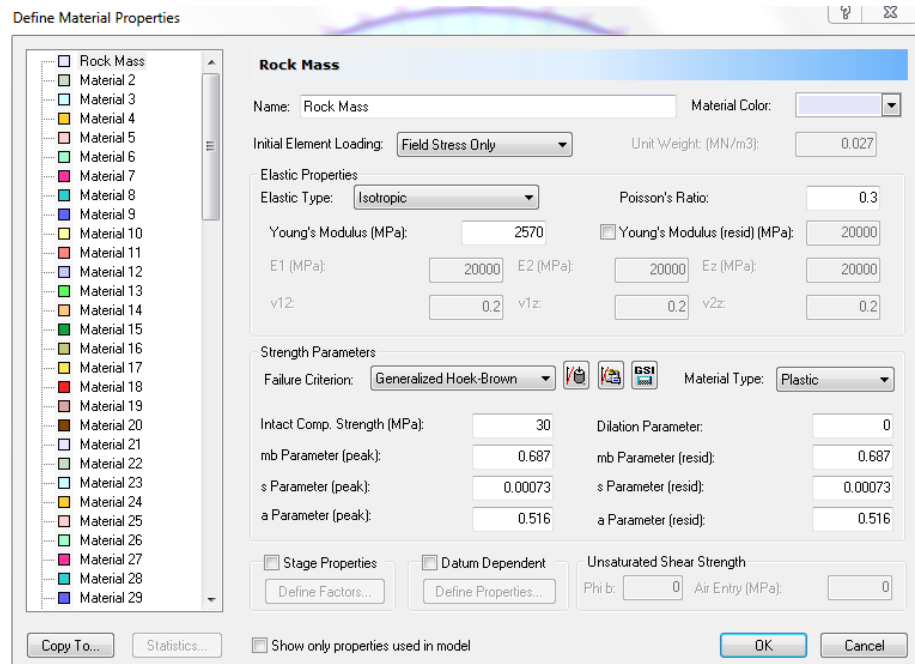
Field Stress Properties	
Field Stress Type:	Constant
Sigma 1 (MPa, Comp. +):	30
Sigma 3 (MPa, Comp. +):	30
Sigma Z (MPa, Comp. +):	30
Angle (degrees from horizontal, CCW):	0
Locked-in horizontal stress (in plane) (MPa, Comp. +):	0
Locked-in horizontal stress (out-of-plane) (MPa, Comp. +):	0

Click OK to close the dialog.

Material Properties



Select **Define Materials** from the **Properties** menu. For Material 1, change the name to Rock Mass. For Initial Element Loading select Field Stress Only. For Young's Modulus enter 2570 MPa and for Poisson's ratio enter 0.3. Under Strength Parameters select Generalized Hoek-Brown for the Failure Criterion. Set the Material Type to be Plastic. Enter the Generalized Hoek-Brown parameters as shown below:



Click OK to close the dialog.

NOTE: the Generalized Hoek-Brown parameters m_b , s , and a , as well as the rock mass modulus, were obtained using the program *RocData*, based on the following Hoek-Brown classification parameters for the rock mass:

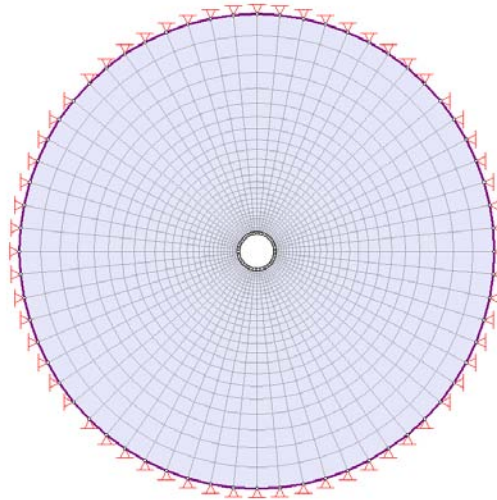
- Sigci (intact uniaxial compressive strength) = 30 MPa
- GSI (Geological Strength Index) = 35
- m_i (intact m parameter) = 7
- D (disturbance factor) = 0

The rock mass modulus was computed from the Simplified Hoek-Diederichs (2005) equation. *RocData* is a program for the analysis of strength data. For more information about *RocData* see the Rocscience website.

Excavation



The tunnel is to be excavated in the first stage so ensure you are looking at Stage 1. From the **Properties** menu select **Assign Properties**. From the Assign Properties dialog, select **Excavate**. Click inside the tunnel. The model should now appear as shown.



TIP: you can also right click inside the tunnel and select **Assign Material** → **Excavate** from the popup menu.

Boundary Pressure

When a tunnel is excavated in three dimensions, the full deformation does not occur immediately at the tunnel face. The rock ahead of the tunnel face begins to deform *before* it is excavated due to the stresses caused by the nearby excavation. As excavation progresses, the boundary will continue to deform as the tunnel face moves away and stresses continue to change. In general, a tunnel does not reach its “two-dimensional” state of deformation until the tunnel face is several diameters away. This is shown schematically below.

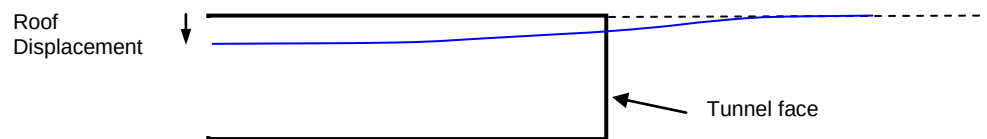
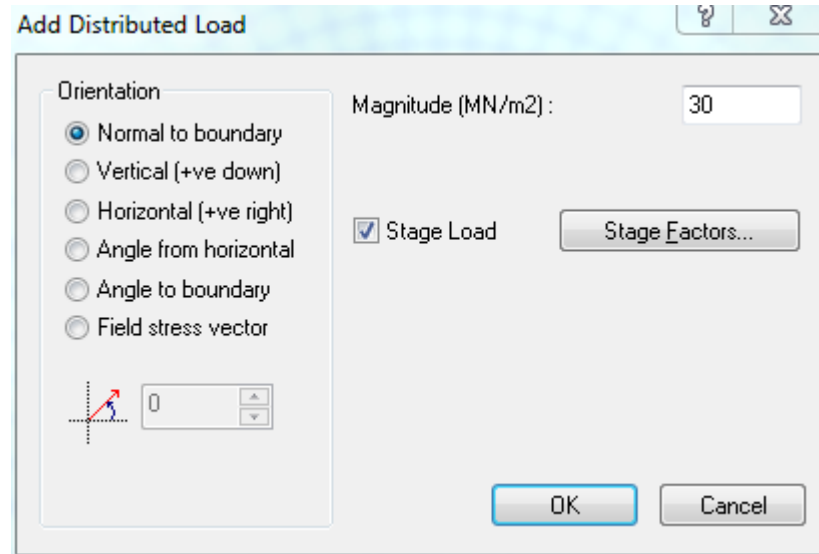


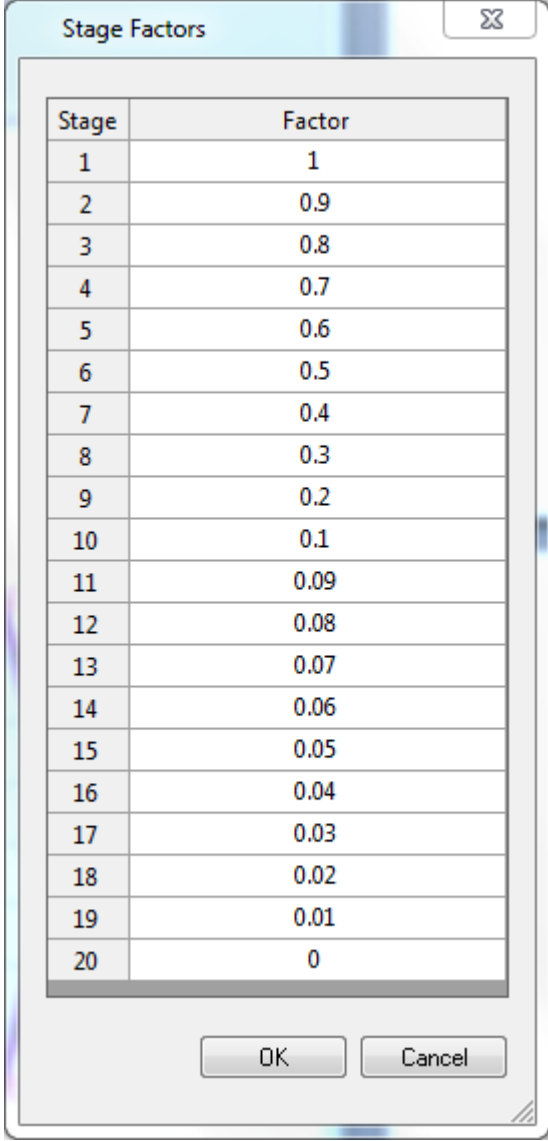
Figure 5. Schematic diagram showing the side view of a tunnel and the roof displacement.



To simulate this effect on the 2-dimensional model, we will apply a traction (distributed load) to the inside of the tunnel boundary, that gradually decreases through the stages. Ensure you are looking at Stage 1. Go to **Loading** → **Distributed Loads** → **Add Uniform Load**. Ensure that the Orientation is Normal to boundary, and enter 30 MN/m² for the Magnitude. This should exactly balance the field stress so that very little deformation will occur at the initial stage.



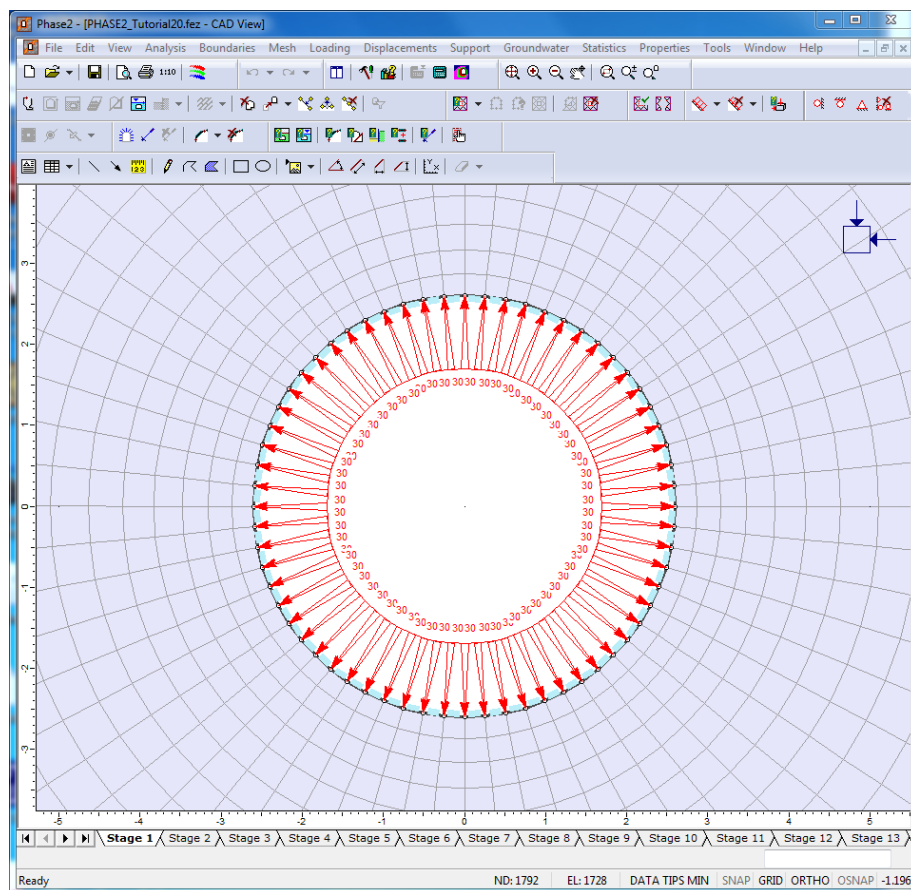
Now click on the **Stage Load** check box as shown above, and select the Stage Factors button. You will now see a dialog asking for the Stage Factor for each stage. The stage factor is multiplied by the initial magnitude to get the actual load magnitude for each stage (e.g., in this case a factor of 1 will apply a load of 30 MN/m² and a factor of 0.5 will apply 15 MN/m²). We wish to gradually decrease the applied load so enter the values for each stage as shown below.



The Stage Factors dialog box contains a table with 20 rows. The first column is labeled 'Stage' and the second column is labeled 'Factor'. The factors decrease linearly from 1.0 at stage 1 to 0.0 at stage 20. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

Stage	Factor
1	1
2	0.9
3	0.8
4	0.7
5	0.6
6	0.5
7	0.4
8	0.3
9	0.2
10	0.1
11	0.09
12	0.08
13	0.07
14	0.06
15	0.05
16	0.04
17	0.03
18	0.02
19	0.01
20	0

Click OK to close the Stage Factors dialog and click OK to close the Add Distributed Load dialog. You will now be asked to select the boundary segments on which to apply the load. Select all of the segments that make up the tunnel by using a selection window. Click somewhere above and to the left of the tunnel and hold down the left mouse button. Drag a window to encompass the entire tunnel and release the mouse button. Hit Enter (or right click and choose Done). You should now see the distributed load applied to the inside of the tunnel as shown (you may need to zoom in).



Click through the stages 1 to 20 to ensure the load is decreasing. Since we entered a stage factor of 0 for stage 20, there should be no load displayed for the final stage.

TIP: to quickly view different stages, you can select the **Page Down** or **Page Up** keys to increase or decrease the stage.

Liner Properties

Now let's define the properties of the steel set liner. For the first part of this tutorial we will define a steel-set liner with no sliding gap.



From the **Properties** menu select **Define Liners**. Change the name of Liner 1 to Steel Set. Change the Young's Modulus to 3133 MPa and the thickness to 0.24 m. Leave all other default values. Click OK to close the dialog.

NOTE:

- **IMPORTANT!!!** The liner properties which we enter (Young's Modulus and Thickness) are actually the properties of an **equivalent** liner of uniform cross-section, which has the same elastic response as a series of equally spaced steel sets. See below for more information.
- Also note that the liner is Elastic. This means that the liner will not fail, it will respond elastically regardless of the stress applied to the liner. (To consider liner failure, we would need to set the liner Material Type = Plastic, and enter the liner Compressive and Tensile Strength. This is beyond the scope of this tutorial, and is left as an optional exercise to explore after completing this tutorial).

The screenshot shows the 'Define Liner Properties' dialog box for a 'Steel Set' liner. The dialog is titled 'Define Liner Properties' and has a list of liners on the left: Steel Set, Liner 2, Liner 3, Liner 4, Liner 5, Liner 6, Liner 7, Liner 8, Liner 9, Liner 10, Liner 11, Liner 12, Liner 13, Liner 14, Liner 15, Liner 16, Liner 17, Liner 18, Liner 19, Liner 20, Liner 21, Liner 22, Liner 23, Liner 24, and Liner 25. The 'Steel Set' liner is selected. The main panel shows the following settings:

- Name:** Steel Set
- Color:** Blue
- Liner Type:** Standard Beam
- Elastic Properties:**
 - Young's Modulus (MPa): 3133
 - Poisson's Ratio: 0.2
- Strength Parameters:**
 - Material Type: ☒ Elastic ☐ Plastic
 - Compressive Strength (peak) (MPa): 35
 - Compressive Strength (residual) (MPa): 5
 - Tensile Strength (peak) (MPa): 5
 - Tensile Strength (residual) (MPa): 0
- Geometry:**
 - ☒ Thickness (m): 0.24
 - ☐ Area (m²): 0.1
 - Moment of Inertia (m⁴): 8.3e-005
- Include Weight in Analysis:**
 - Unit Weight (MN/m³): 0.02
- Pre-Tensioning:**
 - Pre-Tensioning Force (MN): 0
- Sliding Gap:**
 - ☐ Sliding Gap
 - Strain at Locking: 6 %
- Beam Element Formulation:** Timoshenko

At the bottom, there is a 'Copy To...' button, a checkbox for 'Show only properties used in model', and 'OK' and 'Cancel' buttons.

Equivalent Uniform Liner Properties

It is very important to understand how we arrived at the liner properties (Young's Modulus and Thickness) for this example. When we have a liner which is composed of a series of discrete, equally spaced support elements (in this case, steel sets), then to define the equivalent liner properties in *Phase2*, we can define the properties of a liner of uniform cross-section which is equivalent in behaviour to the actual liner system.

For details about how these equivalent liner properties can be calculated, see the Theory section in the *Phase2* Help system (see **Theory > Liners > Equivalent Properties for Steel Set and Shotcrete Liners**). This document describes how to obtain equivalent liner properties for multi-component liners such as steel set and shotcrete support systems. The same equations can also be used to determine the equivalent uniform properties of a steel-set only support system, by setting the modulus of one of the liner components to zero.

For this example, the modulus and thickness of the equivalent uniform liner section is derived from a support system consisting of CP 160 steel sets spaced at 1.5 meters along the length of the tunnel, with 0.2 meter thick shotcrete in between the steel sets. (NOTE: in this tutorial we have not discussed the shotcrete component of the liner, however in practice the steel set rings would normally be augmented by shotcrete support).

Add Liner

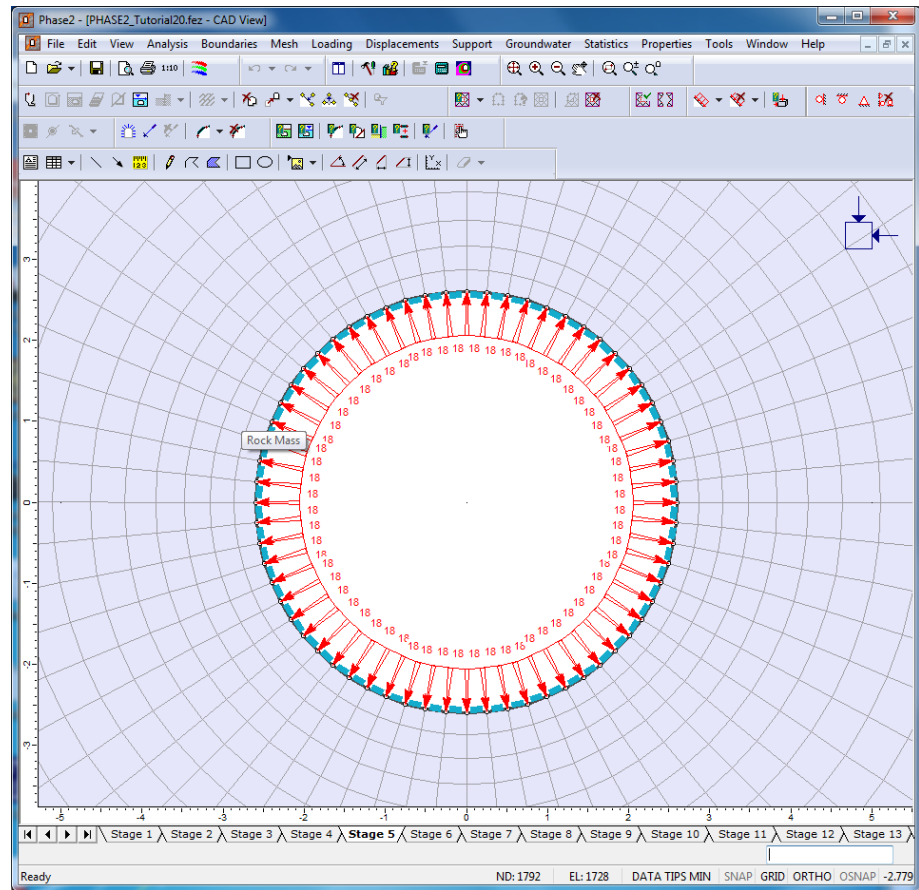
We will now add the liner. We need to determine the stage at which the liner will be applied. Remember that the staging in this model is intended to simulate the 3-dimensional advance of the tunnel face, by gradually decreasing the applied load on the tunnel boundary.

One could perform a true three-dimensional or axisymmetric analysis to determine the state of stress and deformation near the tunnel face and therefore accurately determine the stage (i.e. deformation) at which the support should be added.

For this example we will estimate that the deformation at the face is between one quarter and one third of the final deformation. However, the liner cannot be installed immediately at the face. The deformation at which the support is installed is approximately one third to one half of the final deformation. This corresponds to Stage 5 (i.e. 60% applied traction).



Click on the tab for Stage 5 and from the **Support** menu select **Add Liner**. Ensure the chosen liner is Steel Set and the application stage is Stage 5 and click OK. Select the entire tunnel with a selection window. Hit enter to finish entering boundary segments. The model for Stage 5 should now look like this:



You have now completed the modeling. Select **Save As** from the **File** menu and save the model.

Compute



Run the model by pressing the **Compute** button on the toolbar. The analysis will take a couple of minutes to run.

Once the model has finished computing (Compute dialog closes), click the **Interpret** button to view the results.

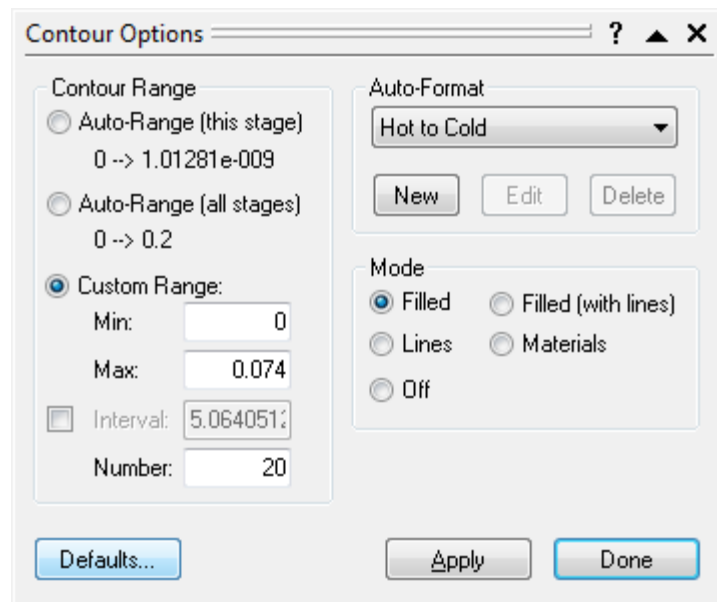
Interpret (no sliding gap)



After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You will see a screen showing the maximum compressive stress for Stage 1. Since we specified the initial traction inside the tunnel to exactly balance the field stress, you should see a constant stress of 30 MPa throughout the rock mass.

Now plot displacements by selecting **Total Displacement** from the drop-down menu on the toolbar. All displacements should be 0 at Stage 1.

To facilitate comparison between stages we should fix the contour range to be the same for all stages. To do this, right click on the model and select **Contour Options**. Select Custom Range and set the maximum to 0.074 (7.4 cm) as shown.



Click Apply and then Done.



Now click through the remaining stages. You will see the displacement around the tunnel increasing as the applied tractions decrease. If you turn on the displacement vectors by clicking the Display Deformation Vectors button, you will see that all of the deformation is radially inwards. The displacements for Stage 10 are shown below.

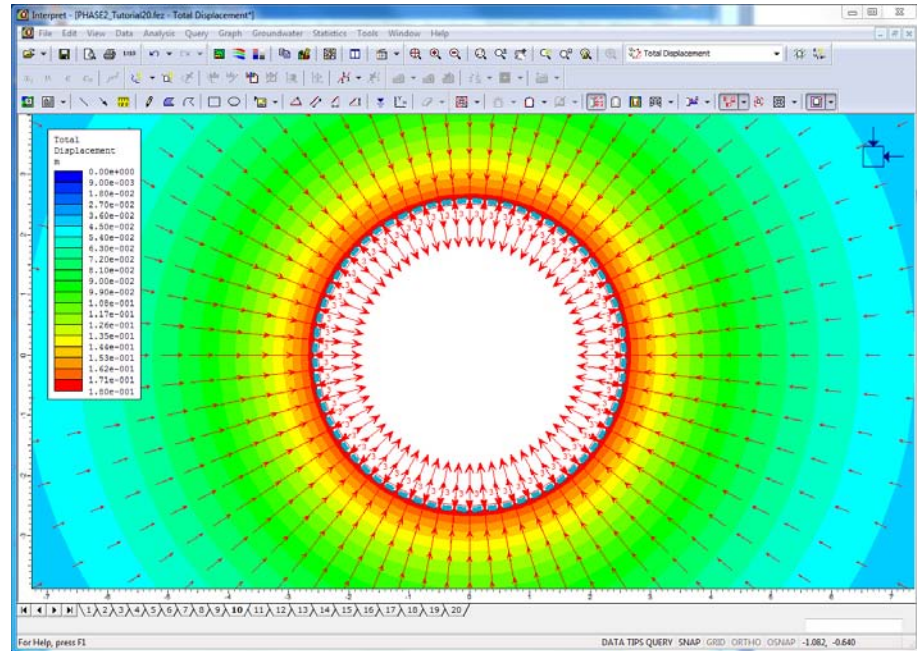


Figure 6. Displacements, Stage 10, no sliding gap.

It is interesting to plot the displacement through the stages. From the **Graph** menu select **Graph Single Point vs. Stage**. Select any point on the tunnel boundary. In the resulting dialog ensure that the Vertical Axis is Query Data and the Horizontal Axis is Stage Number. Click Create Plot. You will see a graph as shown.

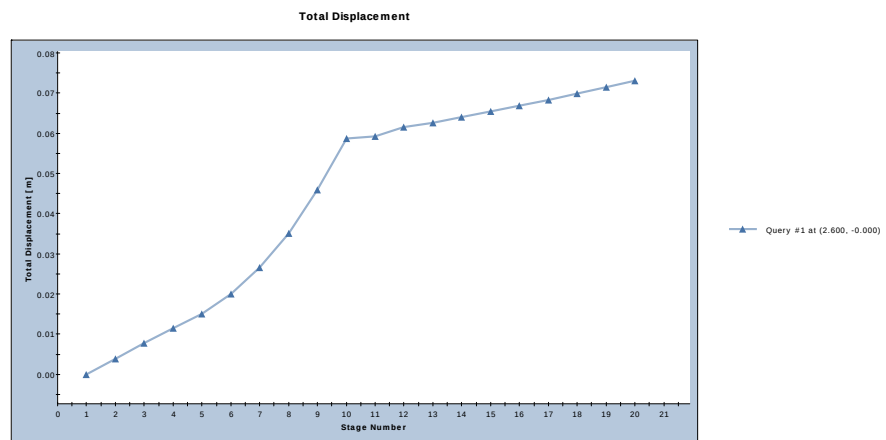


Figure 7. Displacement versus stage number.

You can see the displacement increasing through the stages. The change in slope after stage 10 is due to the fact that we changed the rate at which applied tractions decreased after Stage 10 (from 10% per Stage to 1% per stage).



Interestingly, after stage 5 there appears to be an increase in the rate of displacement even though we installed the liner at Stage 5. This is because after Stage 5 the rock begins to fail. You can observe this by going back to the plot of the model, turning off the displacement vectors and turning on the failed elements (click the **Display Yielded Elements** button). The model in Stage 6 is shown below.

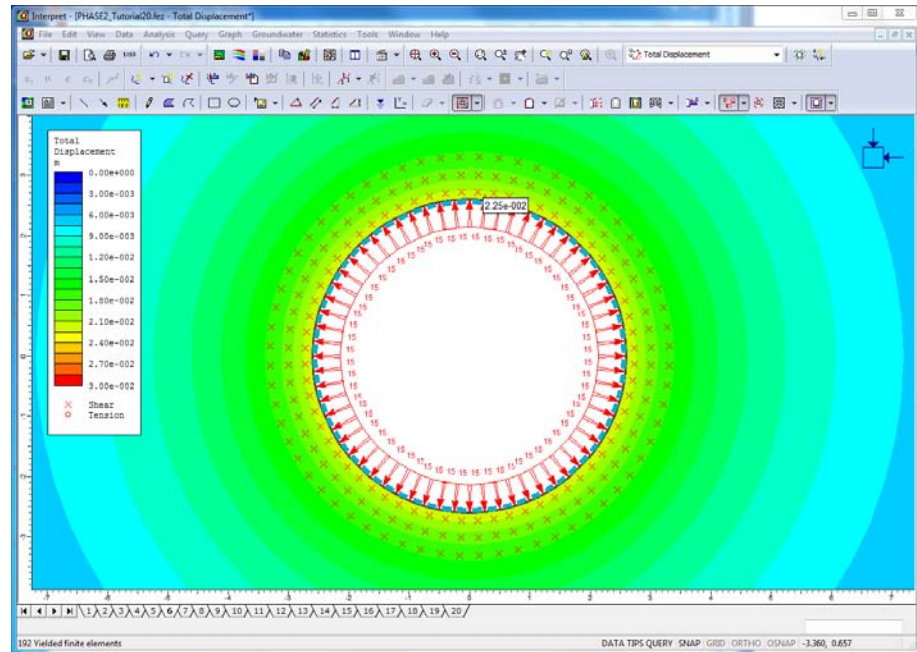


Figure 8. Yielded elements, Stage 6, no sliding gap.

Note that even though some failure occurs in this model, significantly more failure would occur if the liner were not in place (try rerunning the model without the liner. You will see massive failure around the tunnel and displacements > 50 cm by Stage 20).

Liner Loading

Now click on the tab for Stage 5. Recall that we installed the liner at Stage 5. The displacements are still increasing but now the liner is starting to take up some of the load. To see this, right click on the liner and select **Show Values** → **Axial Force**. Turn off the yielded elements and your screen should look like this.

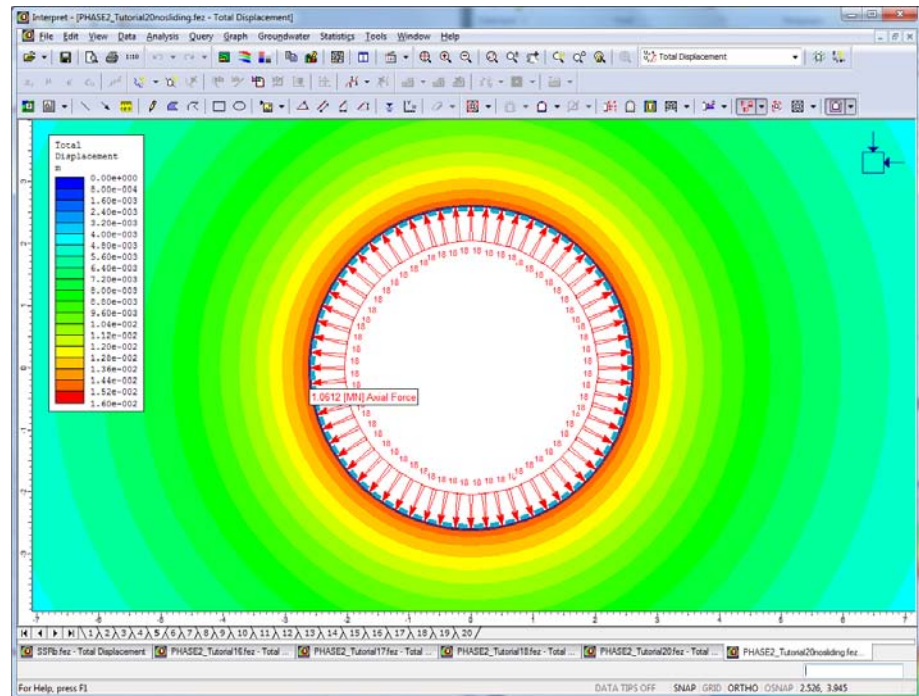


Figure 9. Liner axial force, Stage 5, no sliding gap.

At Stage 5 the liner is taking an axial force of about 1 MN. Note that the bending moments on the liner are zero since the stresses are hydrostatic and the excavation is a circle (you can test this by selecting Bending Moment from the Show Values menu when you right click on the liner).

Click through the subsequent stages. You will see the axial force on the liner increasing with each stage. In stage 20, the axial force on the liner is about 18.5 MN. This force is likely large enough to cause failure of the support system.

This rough analysis suggests that the liner is likely to fail if installed as a rigid entity without sliding gaps. We will now run the model using a liner with sliding gaps to try to decrease the axial stress in the liner while still maintaining the integrity of the tunnel.

Model (with sliding gap)

Go back to the *Phase2* Model program. Open the saved file from the previous part of this tutorial if necessary. We will use the same model as before but now we will include a sliding gap in the liner and observe the effect.

Liner Properties



Select **Define Liners** from the **Properties** menu. At the bottom right of the dialog select the **Sliding Gap** checkbox.

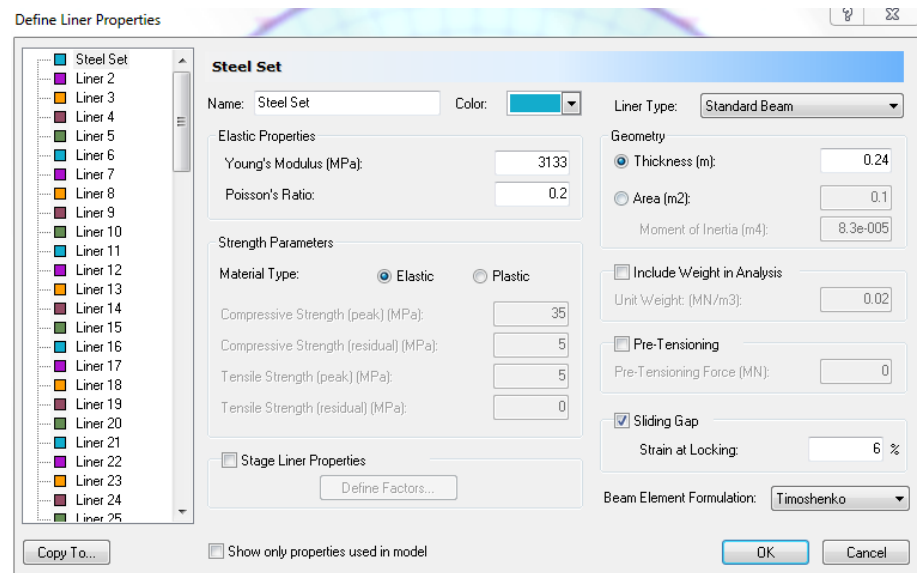


Figure 10. Sliding Gap option in Define Liner Properties dialog.

Enabling the **Sliding Gap** option means that:

1. There will be no axial force in the liner until the locking strain has been reached.
2. However, the liner can resist bending moments before the locking strain has been reached. So in general, the bending moment can be non-zero even prior to locking.

The point at which locking occurs is determined from the Strain at Locking value. The definition of the **Strain at Locking** value in *Phase2* is as follows.

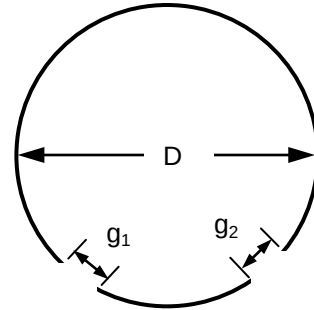
Calculating Strain at Locking for a Liner with Sliding Gaps

The strain at locking refers to the circumferential strain that the liner goes through after installation to the point of locking. This strain is calculated by:

$$\frac{\Delta L}{L} = \frac{\Sigma(\text{gap lengths})}{\text{initial circumference}}$$

For a steel set with two sliding gaps as shown, the strain at locking is calculated as follows:

$$\frac{\Delta L}{L} = \frac{g_1 + g_2}{\pi D} \times 100\%$$



In our model, we will assume that we have two gaps of 0.5 m each (see Figure 1 and Figure 3). The diameter of the tunnel is 5.2 m, therefore the value for Strain at Locking is ~6% by the equation above. Enter this value in the dialog as shown.

Click OK to close the dialog.

Sliding Gap Location

It is important to note the following:

- In the actual steel set support system, the sliding gap(s) are located at certain positions on the circumference of the liner. Typically, 2 or more sliding gaps will be used around the circumference.
- In the *Phase2* model, the liner sliding gap *does not have any specific physical location along the liner*. Locking occurs when the total average strain along the liner is equal to the locking strain. There is no actual physical location to the liner sliding gap(s).

Your modified model is now finished. Save your model by choosing Save As from the File menu.

Compute



Run the model by pressing the **Compute** button on the toolbar. The analysis should take a couple of minutes to run.

Once the model has finished computing (Compute dialog closes), click the Interpret button to view the results.

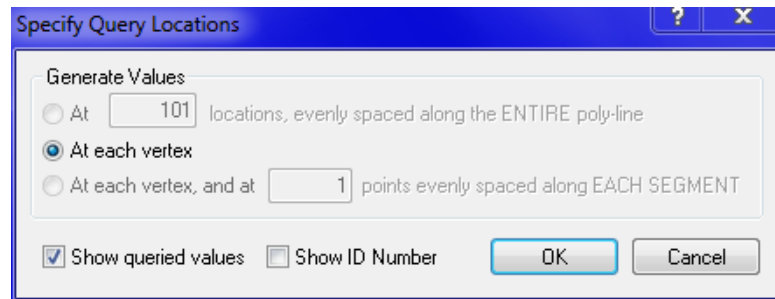
Interpret (with sliding gap)



The model behaviour up until Stage 5 will be the same as before. Select the Stage 5 tab. Change the contours to show Total Displacement. Now right click on the liner and select **Show Values** → **Axial Force**. You will see that the axial force is 0 MN. This is because the liner gaps are sliding and have not yet locked.



It is useful to know the exact roof displacement at each stage. Select **Add Material Query** from the **Query** menu. Click on a point on the tunnel roof and hit Enter. Check the box to Show Queried Values.



Click OK and you will now see the radial displacement at the tunnel roof as shown (1.58 cm).

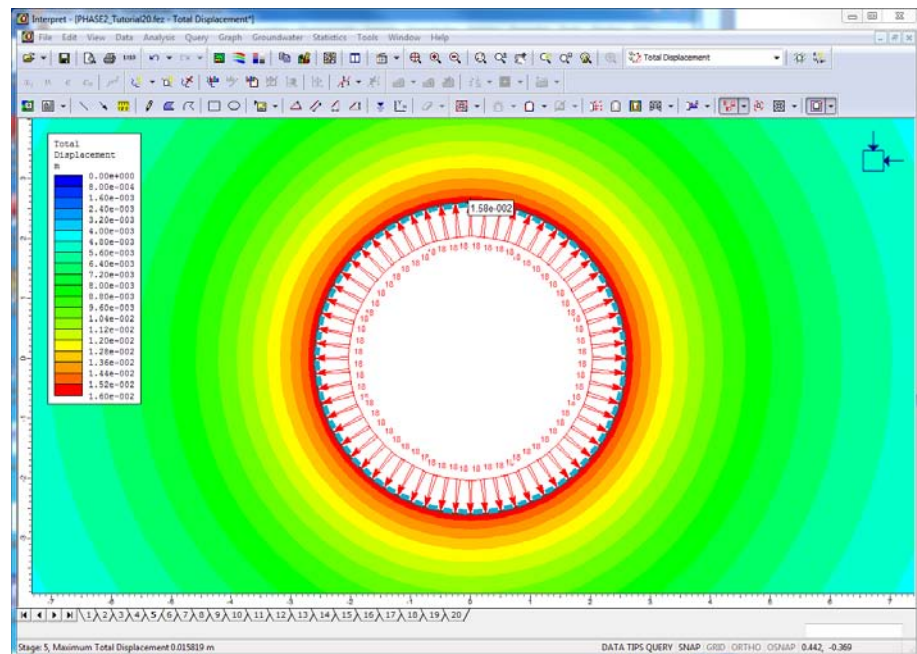


Figure 11. Stage 5 results with sliding gap

If you click through the subsequent stages, you will see that the liner does not start taking axial stress until Stage 11. At this point the displacement at the tunnel roof is 17.7 cm as shown.

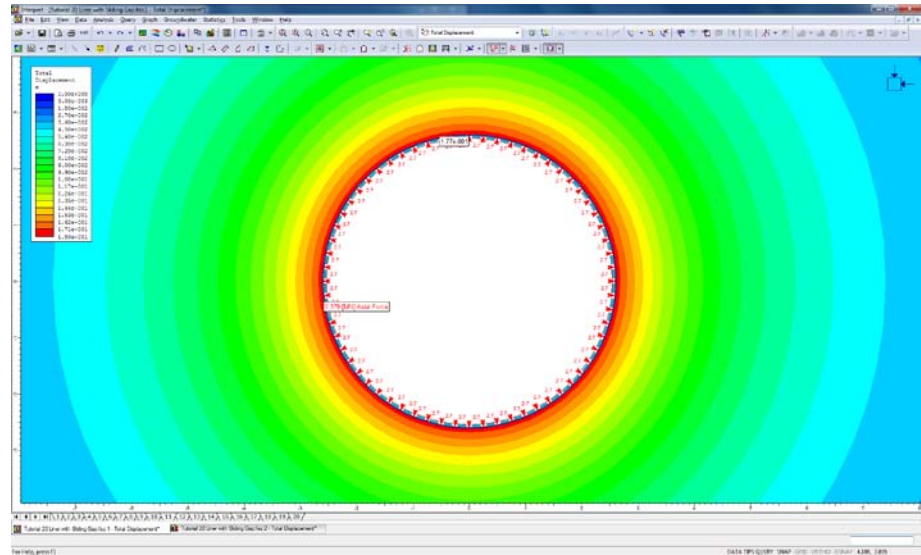


Figure 12. Stage 11 results with sliding gap.

We can use these values to calculate the circumferential strain and check against the Strain at Locking value (6%) that we entered in the Define Liner Properties dialog. To get the circumferential strain we need to know the change in the circumference *from the time of liner installation*. So we need the value of displacement at Stage 5 when the liner was installed (0.0158 m) and at Stage 11 when the liner starts taking load (0.177 m). Assuming that all of the displacement is radial,

$$\begin{aligned}\Delta \text{Circumference} &= \pi \times \Delta \text{Diameter} = 2\pi \times \Delta \text{Radius} \\ &= 2\pi \times (0.179 - 0.0158) \\ &= 1.025 \text{ m}\end{aligned}$$

$$\begin{aligned}\% \text{Strain} &= \frac{\Delta \text{Circumference}}{\text{Circumference}} \times 100\% = \frac{1.025}{5.2\pi} \times 100\% \\ &= 6.27\%\end{aligned}$$

This is very close to the 6% value of circumferential strain specified for the Strain at Locking (the calculated value is slightly higher because some strain has occurred past the point of locking, at Stage 11).

Click through the remaining stages. You will see the displacements increasing as the tractions decrease. The maximum displacement in Stage 20 is ~20 cm. This is higher than that observed for the liner with no sliding gap (7.3 cm) but much less than the displacement that would occur with no liner at all (~50 cm).

Let's compare the effect of the sliding gap on the rock mass yielding.

1. Open the file from the first part of the tutorial (liner with NO sliding gap).
2. Tile the views vertically so that you can easily compare results for the two files, with and without the sliding gap.
3. For each view: select Stage 20, show the Strength Factor contours, display the yielded elements, and use Show Values to display the liner axial force.
4. Your screen should look similar to the following (zoom in or out as required).

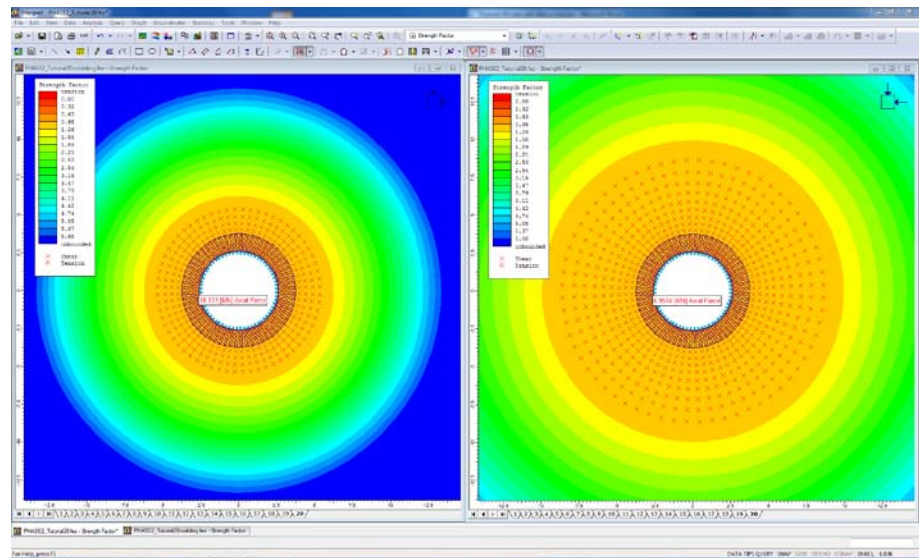


Figure 13. Comparison of rock mass yielding and liner axial force with NO sliding gap (left) and WITH sliding gap (right).

As you can see, including a sliding gap in the liner allows substantially greater failure of the rock mass. However, the final axial force in the liner is much lower. At Stage 20, the liner with sliding gaps is supporting an axial force of 6.4 MN, compared to 18.5 MN with no sliding gap.

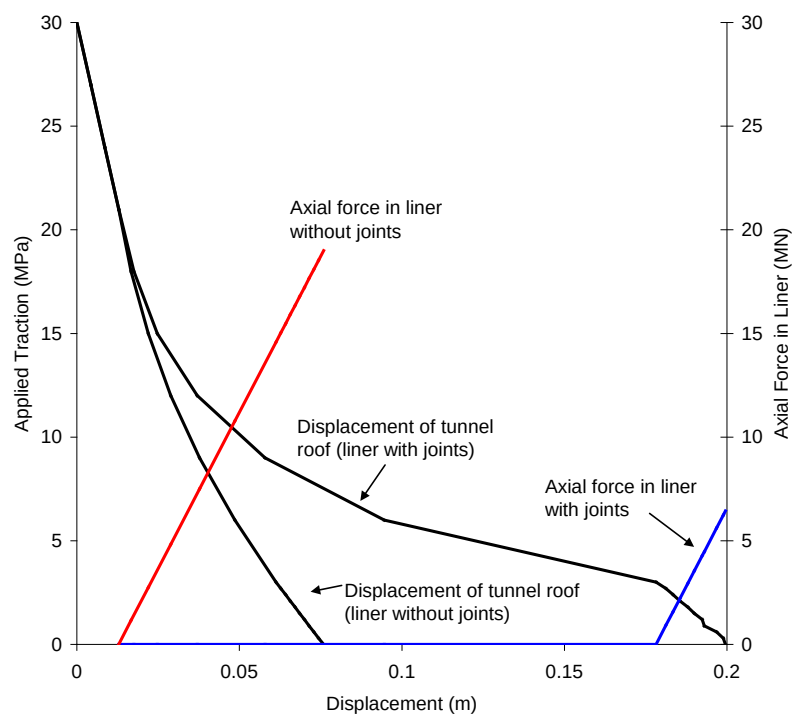
The following table summarizes some key results for the 3 different cases (no liner, liner with no sliding gap, and liner with sliding gap).

	Maximum Radial Displacement (m)	Maximum Axial Force in Liner (MN)	Number of Yielded Elements
No Liner	0.546	n/a	1152
Liner (no sliding gap)	0.073	18.5	512
Liner (with sliding gap)	0.197	6.4	832

This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.

Additional Exercises

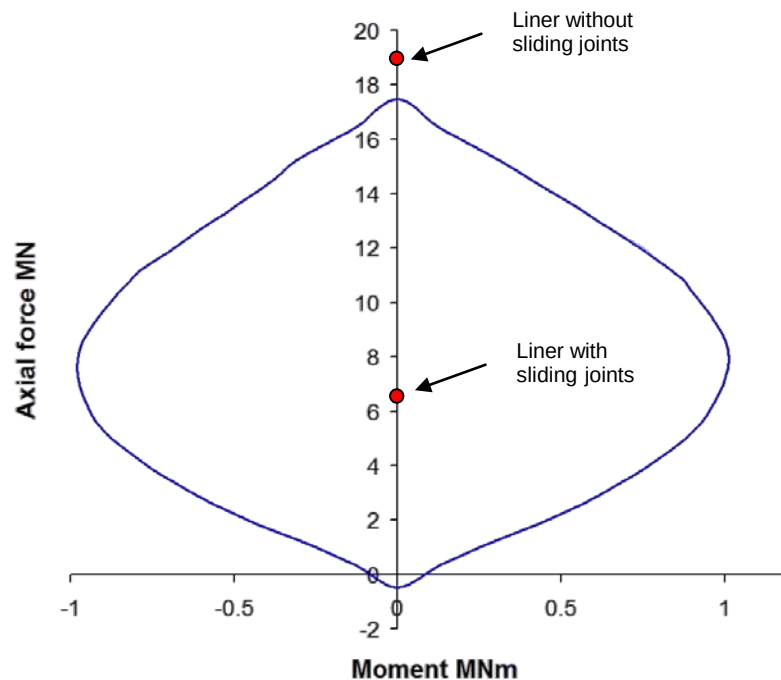
You can create a plot similar to **Figure 2** for the two models by extracting the displacements, tractions and liner axial loads for each stage. Plot the applied tractions and liner axial forces versus displacements. The plot should appear as shown.



This is not exactly the same as **Figure 2** since we don't know the support pressure provided by the liner (only the axial forces); however the plot is still informative. Notice how the displacement curve for the model with sliding gap turns sharply downwards when locking is achieved and the liner begins to take on axial load. Since the curve is becoming quite flat at this point, it is clear that the liner is preventing significant further deformation without a large amount of extra support pressure.

Liner Capacity Envelopes

If we know the specifications for the capacity of our support system, we can check if the force on the liner falls within acceptable limits. A program that can generate such capacity diagrams is Response 2000 (<http://www.ecf.utoronto.ca/~bentz/r2k.htm>). The capacity envelope for the liner considered in this tutorial is shown below.



Points lying inside the blue envelope are acceptable, whereas states outside the envelope represent potential failure of the support system. In our models, the bending moments are zero since the stresses are hydrostatic. The liner forces for the two models are plotted on the diagram. According to this diagram, the force on the liner without sliding gaps will result in failure of the liner, whereas the liner with sliding gaps is safely within the design limits.

The following reference provides a useful overview and discussion of liner capacity envelopes:

- Sauer, G., Gall, V., Bauer, E and Dietmaier, P. 1994. Design of tunnel concrete linings using capacity limit curves. in *Computer Methods and Advances in Geomechanics*, Eds.: Siriwardane & Zaman, page 2621 - 2626 Rotterdam, NL.